

Irrigation inefficiency and water distribution losses as drivers of Kura River depletion and agricultural resource inefficiency in Azerbaijan

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Keywords

agricultural water use, green economy, irrigation efficiency, Kura River, water distribution losses

ABSTRACT

The Kura River is a strategic water source for Azerbaijan, where irrigated agriculture depends heavily on reliable conveyance and efficient field application. This study integrates two independently prepared research works on Kura River depletion and agricultural water-use problems and places their observations within a resource-efficiency and green-economy framework. The source materials identify excessive irrigation withdrawals, leakage from aging distribution networks, widespread use of conventional surface irrigation, drainage and salinity problems, and weak coordination of water management as major drivers of water loss and agricultural vulnerability. A structured qualitative synthesis was combined with verification against recent World Bank, FAO, national policy and peer-reviewed sources. The World Bank's 2024 water-security assessment estimates average irrigation water received at about 7.03 billion m³ per year against crop water consumption of about 2.33 billion m³, implying a potential national-level difference of around 67%, while overall irrigation-system efficiency was estimated at 37%. These figures support the central argument of the original research that losses occur both during conveyance and at farm level. Economically, such inefficiency raises the amount of water that must be abstracted and delivered, increases exposure to shortages, and can reduce farm productivity through poor timing, salinity and waterlogging. Modernized canals, metering, drip and sprinkler systems, improved drainage and allocation planning are therefore not only technical measures but green-economy investments in water productivity, agricultural resilience and natural-capital protection.

INTRODUCTION

Agricultural production in Azerbaijan is strongly linked to irrigation, especially in the Kura–Araz lowland and other dry or seasonally water-limited areas. The Kura River supports irrigation, municipal supply, energy and ecosystems, yet the same river is subject to growing pressure from climate variability, abstraction and inefficient distribution. For agriculture, the central management problem is not only how much water is available at the river but how much of the withdrawn water reaches fields at the right time and is used productively.

The two source research works used in this article approach the issue from complementary directions. Amrullayeva examined the non-systematic use of the Kura River, emphasizing agricultural overuse, distribution leaks and the absence of fully coordinated basin governance. Shahpalangli focused on agricultural water use and identified open-channel seepage, evaporation, traditional surface irrigation, poor drainage and salinity as constraints that lower the effectiveness of irrigation. Both works proposed modernization through canal rehabilitation, drip and sprinkler irrigation, better scheduling and stronger water management.

These concerns have direct economic significance. Water lost before reaching crops still requires abstraction, storage, conveyance and management, while unreliable supply can reduce yields or force farmers to irrigate at suboptimal times. Inefficient irrigation can also raise groundwater levels and salinity, converting a water-management

problem into land degradation and future production costs. In a green-economy framework, reducing these losses means increasing output and resilience per unit of natural resource rather than simply expanding water withdrawals.

The objective of this study is to synthesize the original research materials into an academic assessment of irrigation inefficiency and distribution losses as drivers of Kura River depletion and agricultural resource inefficiency. The study asks three questions: (1) where do the principal irrigation losses occur; (2) how do those losses affect agricultural and environmental performance; and (3) which modernization measures offer the strongest combined water-security, economic and green-economy benefits? The source projects did not conduct farm-level flow measurements or a cost-benefit survey; therefore, quantitative values used in the discussion are drawn only from traceable published sources.

LITERATURE REVIEW

Recent national evidence confirms that irrigation inefficiency is a structural water-security issue. The World Bank (2024) estimates that average irrigated land in Azerbaijan is about 1.02 million ha compared with an average irrigable area of 1.54 million ha. Its Water Accounting Tool assessment reports average crop water consumption of approximately 2.33 billion m³ per year against about 7.03 billion m³ of irrigation water received, potentially indicating losses of roughly 67% at national level. The same assessment estimates overall

irrigation-system efficiency at 37%, with major losses attributed to leakage and seepage from aging infrastructure.

A 2026 agricultural-economics study reaches a similar conclusion. Mustafayeva et al. (2026) identify water losses in agricultural conveyance as a major constraint on sectoral efficiency and argue that improving irrigation infrastructure and water-saving technologies is central to agricultural water security. The study frames the problem economically: scarce water should be managed so that the value generated per unit of water rises rather than relying on higher withdrawals to compensate for inefficient delivery.

The physical consequences of irrigation inefficiency extend beyond water volume. Han et al. (2024) show that water management and drainage conditions are closely linked to soil salinity and agricultural productivity in the Kur-Araz lowland. Their analysis demonstrates that salinity dynamics can materially affect the sustainability of agricultural production. This supports the concern in the source projects that seepage, high groundwater and poor drainage can convert conveyance losses into soil-quality losses.

Water-saving modernization is also embedded in national policy. Azerbaijan's National Strategy on the Efficient Use of Water Resources, approved in October 2024, sets priorities for integrated management, efficient water use, improved irrigation-water provision and water security (President of the Republic of Azerbaijan, 2024). In 2026, implementation responsibilities were further updated, showing that water-sector reform remains active. The strategy provides a policy context for canal rehabilitation, monitoring, efficient irrigation technologies and improved allocation.

At basin scale, technical efficiency also depends on governance. Suleymanov (2025) emphasizes the importance of stronger institutional arrangements for the Kura-Aras basin, while Amrullayeva's original literature review likewise identified fragmented management as a factor that can complicate allocation and planning. FAO AQUASTAT further notes that irrigation withdrawals commonly exceed crop water requirements because losses occur during distribution and application, reinforcing the need to distinguish between water diverted from a source and water productively consumed by crops (FAO, 2025).

MATERIALS AND METHODS

The study used an integrative qualitative design based on two author-prepared research dossiers completed in 2026. Amrullayeva's work was a literature-based investigation of Kura River depletion, agricultural overuse, transboundary management and distribution leakage. Shahpalangli's work examined agricultural water use, open-channel losses, modern irrigation methods, drainage, salinity and farmer-oriented recommendations. Their original themes, terminology and proposed solutions were retained as the foundation of the synthesis.

The source material was coded into five analytical categories: water abstraction and agricultural demand; conveyance and distribution losses; field-level irrigation efficiency; soil and drainage consequences; and economic and green-economy implications. Claims with numerical values were checked against traceable institutional or peer-reviewed sources. Where the original works reported percentages without a verifiable underlying dataset, those values were not reproduced as measured national statistics.

External verification used the World Bank's 2024 Azerbaijan water-security report, the 2024 National Strategy on the Efficient Use of Water Resources, FAO AQUASTAT materials, recent peer-reviewed work on soil salinity and agricultural water efficiency, and the Kura-Aras basin governance literature already cited by

Amrullayeva. The study does not estimate causal effects, farm profits or payback periods for irrigation equipment. The economic assessment is interpretive and focuses on resource productivity, avoided losses, production risk and resilience.

RESULTS AND DISCUSSION

Conveyance losses and the scale of agricultural water inefficiency

Both source works identify the distribution network as a critical point of loss. Amrullayeva describes leakage from poorly maintained channels and pipes, while Shahpalangli emphasizes seepage from open canals and evaporation before water reaches crops. The national-scale evidence is consistent with this diagnosis. World Bank (2024) estimates average irrigation water received at 7.03 billion m³ per year and crop water consumption at 2.33 billion m³, implying a potential difference of about 67%. The report cautions that the difference reflects system-wide losses and accounting limitations rather than a single measured leakage rate, but it nonetheless demonstrates a large efficiency gap.

The same assessment estimates national irrigation-system efficiency at approximately 37%, with substantial regional variation. This is economically important because low conveyance and field efficiency increase the amount of water that must be diverted to deliver a given amount of effective crop water. In a water-scarce system, this raises competition between users and reduces the buffer available during drought or low-flow periods. The source projects' recommendation to rehabilitate canals and improve distribution is therefore strongly supported by recent evidence.

Figure 1 compares average irrigation water received with estimated crop water consumption in the World Bank assessment. The figure should be read as a resource-efficiency indicator, not as a direct financial-loss calculation. The difference includes conveyance losses, field application losses, return flows, measurement uncertainty and other system effects.

Table 1 links the main loss mechanisms identified in the source research to their agricultural, economic and green-economy implications.

Field application, crop productivity and the economics of water-saving technology

Traditional surface irrigation can deliver water unevenly and may require larger withdrawals than localized systems when management is weak. Shahpalangli's work therefore prioritizes drip and sprinkler irrigation. These technologies can improve control over timing and application, particularly when combined with soil-moisture monitoring, pressure regulation and appropriate crop scheduling. Their value is not simply the physical volume of water saved; it is the possibility of producing more reliable output with less water diverted from rivers and reservoirs.

The green-economy benefit arises from higher resource productivity. Water-saving technology can reduce exposure to shortages, support higher-value production where technically suitable and lower the need for continual expansion of supply infrastructure. However, modernization also requires capital, maintenance, filtration, energy and technical skills. For that reason, investment decisions should be crop- and location-specific. The objective is not to replace every surface-irrigation system but to target the sections of the network and farms where the economic value of reduced losses is greatest.

Azerbaijan's 2024 National Strategy supports this direction by prioritizing efficient use, irrigation reliability and integrated management. From an optimization perspective, the policy challenge

is to allocate rehabilitation funds between main canals, secondary networks, on-farm systems, drainage and digital monitoring so that marginal water savings and production benefits are maximized.

Table 1. Irrigation inefficiency pathways and green-economy responses in the Kura-dependent agricultural system

Inefficiency pathway	Green-economy response
Aging or unlined conveyance network	Canal lining or piping, rehabilitation and metering
Poorly controlled surface irrigation	Drip or sprinkler systems where suitable and soil-moisture scheduling
Weak drainage and high groundwater	Drainage maintenance and groundwater monitoring
Limited measurement and allocation data	Digital monitoring and seasonal water accounting
Fragmented basin and sectoral coordination	Integrated governance and coordinated allocation

Seepage, groundwater and salinity as hidden production costs

Conveyance loss is not always a simple disappearance of water; some seepage can recharge groundwater. In poorly drained agricultural areas, however, excessive recharge can raise the water table and contribute to salinity. Amrullayeva's source work explicitly links leakage, rising groundwater and salinization, while Shahpalangli emphasizes drainage maintenance and careful scheduling. Han et al. (2024) provide peer-reviewed evidence that water management and drainage conditions are important determinants of soil salinity and agricultural productivity in the Kur-Araz lowland.

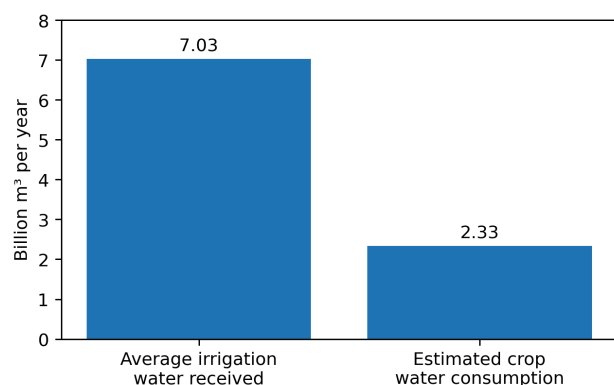


Figure 1. Average irrigation water received and estimated crop water consumption in Azerbaijan based on World Bank (2024)

Salinization has a clear economic pathway: degraded soils may require additional drainage, leaching, land reclamation or crop substitution and can reduce the productivity of existing farmland. This means that investments in canal rehabilitation and drainage can yield joint benefits by conserving water and protecting land capital. A green-economy strategy should therefore assess water efficiency together with soil condition rather than treating irrigation and land degradation as separate policy fields.

Integrated modernization priorities for agricultural water security

The combined evidence supports a phased modernization package. First, high-loss conveyance sections should be identified through metering, water accounting and condition surveys, followed by lining, piping or targeted rehabilitation where justified. Second, drip or sprinkler systems should be expanded where crop value, soil conditions and farm scale support their operation. Third, drainage and groundwater monitoring should be maintained to prevent water-saving interventions from creating new salinity risks. Fourth, water delivery schedules should reflect crop demand rather than fixed routine allocation. Fifth, digital measurement and transparent water

accounting should connect river abstraction, canal delivery and field use.

Economic resilience, governance and study limitations

The economic case for improving irrigation efficiency is strongest when water scarcity, infrastructure deterioration and agricultural dependence occur together. Reduced distribution losses can lower the volume that must be abstracted for the same productive use, protect a larger share of river flow for other sectors and ecosystems, and improve the reliability of farm supply. At basin scale, stronger institutional coordination can also reduce uncertainty in allocation and improve planning, as emphasized by Suleymanov (2025). These outcomes fit a green-economy model in which infrastructure investment increases productivity while reducing pressure on natural capital.

The study has limitations. The two underlying research works are qualitative educational dossiers rather than standardized hydraulic audits. They contain useful problem identification and practical observations but no direct measurements of canal seepage, farm-level application efficiency, crop yield response or investment cost. Some percentages in the original Shahpalangli material were not linked to traceable datasets and were therefore excluded from the quantitative results. Future research should combine canal flow measurements, farm water accounts, remote sensing, soil salinity data and cost-benefit analysis so that irrigation investments can be prioritized according to both water savings and economic returns.

CONCLUSIONS

This study integrates two original research works into a unified assessment of how irrigation inefficiency and water-distribution losses contribute to Kura River depletion and agricultural resource inefficiency in Azerbaijan. The source research correctly identifies conveyance leakage, traditional irrigation, drainage constraints and weak management as interconnected pressures. Recent evidence strengthens that diagnosis: the World Bank's 2024 assessment estimates a large gap between irrigation water received and crop water consumption and an overall irrigation-system efficiency of about 37%. The implication is both environmental and economic. Excessive losses increase the water that must be withdrawn, reduce resilience during scarcity, and can contribute to salinity and declining land productivity. Modernized conveyance, metering, localized irrigation, improved scheduling and drainage management should therefore be treated as green-economy investments that raise water productivity while protecting river and soil natural capital. The next research step should be a farm- and canal-level economic assessment that identifies where each unit of investment produces the greatest combination of water savings, yield stability and ecosystem benefit.

AUTHOR CONTRIBUTIONS

I.A. and H.S. contributed the original research materials, topic development, source collection and preliminary assessment of Kura River use, irrigation losses, agricultural water management and proposed solutions. N.B. contributed supervision, conceptual integration, methodological framing, economic and green-economy interpretation, review and editing. All authors contributed to interpretation, reviewed the final manuscript and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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This research received no external funding.

DATA AVAILABILITY

This study is a qualitative synthesis of two author-prepared research dossiers and published secondary sources. No new farm survey, canal-flow

dataset, crop-yield experiment or financial dataset was generated. Quantitative values used in the article are taken from the cited published sources, while the original descriptive research materials are contained in the authors' project files.

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